

Exact Coalescence and the ROF Semigroup Identity for Total-Variation Flow on a Finite Path

Route-A source-local certificate HCS-C279

1 September 2026

Abstract

We solve the Euclidean total-variation gradient flow on every finite unweighted path. An explicit interpolation of edge flux across each constant block gives a piecewise-affine coalescence algorithm: blocks never split, simultaneous collisions merge jointly, and at most $n - 1$ event times occur.

1 Frozen path flow

Let P_n be the path with vertices $1, \dots, n$, where $n \geq 1$. On $\mathbb{R}^n$ with its Euclidean inner product define

$$(Dx)_i = x_{i+1} - x_i, \quad J(x) = \|Dx\|_1 = \sum_{i=1}^{n-1} |x_{i+1} - x_i|. \quad (1)$$

We study the maximal-monotone clock

$$x'(t) \in -\partial J(x(t)), \quad x(0) = x^0. \quad (2)$$

This freezes unit edge weights and the ordinary vertex norm. It is neither an Euler time step nor a continuum limit. Maximal-monotone semigroups provide the general existence framework [1]; graph total-variation flow and its finite-time stabilization are treated in a broader setting by Mazón, Solera, and Toledo [4].

For $a \in \mathbb{R}$, let $\text{Sgn}(a) = \{\text{sign } a\}$ if $a \neq 0$ and $\text{Sgn}(0) = [-1, 1]$. The chain rule for (1) gives

$$\partial J(x) = \{D^\top z : z_i \in \text{Sgn}((Dx)_i)\}. \quad (3)$$

Since J is finite, continuous, and convex, ∂J is maximal monotone; there is a unique global absolutely continuous solution of (2).

Write $\bar{x} = n^{-1} \sum_i x_i^0$ and $\rho(t) = \|x(t) - \bar{x}\mathbf{1}\|_2$. Every $D^\top z$ is orthogonal to $\mathbf{1}$, so the mean is preserved. One-homogeneity implies $\langle g, x \rangle = J(x)$ for $g \in \partial J(x)$. With $g = -x'$ and the convex chain rule,

$$\frac{d}{dt} \frac{\rho^2}{2} = -J(x), \quad \frac{d}{dt} J(x) = -\|x'\|_2^2 \quad (4)$$

almost everywhere.

Lemma 1 (Finite consensus). *The solution reaches $\bar{x}\mathbf{1}$ by a time T satisfying*

$$T \leq \sqrt{n} \|x^0 - \bar{x}\mathbf{1}\|_2, \quad (5)$$

and remains there.

Proof. For each i , path telescoping and averaging give

$$|x_i - \bar{x}| \leq \frac{1}{n} \sum_j |x_i - x_j| \leq J(x).$$

Hence $\rho \leq \sqrt{n} J(x)$. While $\rho > 0$, the first identity in (4) yields $\rho' = -J(x)/\rho \leq -1/\sqrt{n}$. Integration proves (5). At a constant vector the minimal subgradient is zero, so uniqueness keeps the state constant. $\square$

2 Exact block dynamics and joint collisions

A maximal constant block $B = [\ell, r]$ has size $m = r - \ell + 1$. Define its exterior signs

$$s_{\ell-1} = \begin{cases} \text{sign}(x_\ell - x_{\ell-1}), & \ell > 1, \\ 0, & \ell = 1, \end{cases} \quad s_r = \begin{cases} \text{sign}(x_{r+1} - x_r), & r < n, \\ 0, & r = n. \end{cases}$$

Lemma 2 (Plateau flux). *On the zero-jump edges internal to B , set*

$$z_{\ell+k-1} = s_{\ell-1} + \frac{k}{m}(s_r - s_{\ell-1}), \quad k = 1, \dots, m-1. \quad (6)$$

Together with the exterior signs, this is an admissible subgradient flux and every coordinate in B has velocity

$$v_B = \frac{s_r - s_{\ell-1}}{m}. \quad (7)$$

Proof. The interpolation (6) stays in $[-1, 1] = \text{Sgn}(0)$, so it is admissible in (3). Consecutive flux differences are all $(s_{\ell-1} - s_r)/m$ on B . Thus $D^\top z$ is constant there and $-D^\top z$ equals (7). The same interpolation uniquely minimizes the sum of squared consecutive flux differences with the two boundary values. Once the exterior nonzero-edge signs are fixed, $\|D^\top z\|_2^2$ is the sum of these contributions over the maximal plateaus, so the global minimization separates blockwise. Thus the local discrete Dirichlet minimizer is exactly the minimal-norm subgradient selected by the right derivative of the flow. $\square$

Theorem 1 (Complete path coalescence atlas). *The solution of (2) is piecewise affine. Between events each maximal block moves with (7). Blocks never split. At an event, every connected string of adjacent blocks whose affine heights agree is replaced by its union; disjoint simultaneous collisions are handled at the same time. Every event lowers the block count, so there are at most $n - 1$ distinct event times, after which consensus holds.*

Proof. While adjacent heights are unequal, all exterior signs and block sizes are fixed. Lemma 2 therefore produces constant velocities and an affine curve satisfying (2); uniqueness identifies it with the semigroup solution. A current plateau has one common velocity and cannot split before the first adjacent equality.

At equality, union each maximal connected string of equal heights and apply Lemma 2 to the union. It again has one velocity; uniqueness forbids a later split. Joint union makes the result independent of any artificial ordering among simultaneous collisions. Each event removes at least one block, proving the event bound. If several blocks remained forever without another equality, their affine solution would contradict finite consensus from Lemma 1; hence the construction closes. $\square$

If current blocks have heights h_j and velocities v_j , the next event is the least positive number among

$$\tau_j = \frac{h_{j+1} - h_j}{v_j - v_{j+1}}.$$

All quantities are rational whenever x^0 is rational. The formula also covers endpoint blocks through the missing-flux value zero. For $n = 1$ or constant data there are no events.

References

- [1] H. Brezis, *Opérateurs maximaux monotones et semi-groupes de contractions dans les espaces de Hilbert*, North-Holland Mathematics Studies 5, 1973, MR0348562.
- [2] L. I. Rudin, S. Osher, and E. Fatemi, *Nonlinear total variation based noise removal algorithms*, Physica D **60** (1992), 259–268, doi:10.1016/0167-2789(92)90242-F.
- [3] C. Kirisits, O. Scherzer, and E. Setterqvist, *Invariant φ -minimal sets and total variation denoising on graphs*, SIAM Journal on Imaging Sciences **12**(4) (2019), 1643–1668, doi:10.1137/19M124126X.
- [4] J. M. Mazón, M. Solera, and J. Toledo, *The total variation flow in metric random walk spaces*, Calculus of Variations and Partial Differential Equations **59** (2020), article 29, doi:10.1007/s00526-019-1684-z.